

Efficient Heuristics for Classical Simulation of Quantum Circuits Using ZX-Calculus

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Abstract

Quantum circuits require validation and testing to ensure that they perform as expected. Classical simulation is a powerful tool for this purpose, but it is limited by the exponential growth of complexity with the number of qubits. Nevertheless, using the ZX-calculus, a graphical language for quantum mechanics, we can attempt to reduce how exponential this growth is. Recent developments have allowed focus to be placed on complexity being dependent on the number of T -gates that are in the circuit. Given t T -gates, if the complexity of the circuit is $O(2^{\alpha t})$, the aim is to reduce the value of α as much as possible.

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I hereby certify that this is entirely
my own work unless otherwise stated.

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Introduction

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1.1 Motivation

Quantum computing, a field rooted in the principles of quantum mechanics, represents a revolutionary approach to computation. Unlike classical computing, which relies on binary operations performed on bits, quantum computing utilizes quantum bits, or qubits, capable of representing and processing information in ways that classical bits cannot. This unique capability of qubits to exist in superpositions and entangle with each other allows quantum computers to perform certain types of calculations exponentially faster than their classical counterparts. If a classical computer were limited by memory and runtime, a quantum computer could potentially bypass these scaling issues - dubbed the "quantum advantage".

If scalable quantum computers were to become a reality, they could solve some of the most challenging problems in computer science much more efficiently than classical computers. One notable example is Shor's algorithm for factoring large numbers, which could break widely-used cryptographic systems such as RSA. Another significant algorithm is Grover's search algorithm, which provides quadratic speed-ups for unstructured search problems.

Classical computations play an essential role in assisting and accelerating the development of quantum computers. Classical simulation is a fundamental aspect of understanding and designing quantum hardware, often serving as the only means to validate these quantum systems [1]. Additionally, quantum algorithms can be implemented on classical simulators for testing and verification purposes while full-fledged quantum computers remain beyond reach (for now). These simulators emulate the behavior of quantum computers, allowing researchers to simulate processes as if running on actual quantum devices.

This thesis aims to present the speedup of classical simulation of quantum circuits in an almost entirely graphical approach. From the introduction of the qubits, to the representation of quantum circuits and linear maps, the underlying linear algebra that governs the operations of quantum computing is represented in a graphical manner. Using ZX-Calculus, a graphical calculus detailed in section 1.3, quantum circuits can be represented as diagrams, allowing for different discoveries to be more easily made when analysing the structure of said diagrams.

1.2 Basic Notation

In classical computing, a bit is the fundamental unit of information, and is denoted by a boolean value $b \in \{0, 1\}$. In quantum computing, the fundamental unit of information is the qubit, which we denote as $|b\rangle$, where $b \in \{0, 1\}$. Diagrammatically, they are represented as follows:

$$\triangleleft_0 \quad \triangleleft_1$$

We consider these the computational or Z-basis states.

Definition 1.2.1 (Z-basis). The Z-basis or computational basis is defined as

$$\{ \triangleleft_0 \text{---}, \triangleleft_1 \text{---} \}$$

▲

So far, the expressive power is still similar to classical computing. However, the power of quantum computing comes from the ability to exist in superpositions of these states. Concretely, the qubit can be represented in a linear combination of the basis states.

Definition 1.2.2 (Pure state). A pure state $|\psi\rangle$ is a state

$$\triangleleft_\psi \text{---} = a \triangleleft_0 \text{---} + b \triangleleft_1 \text{---}$$

where $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$.

▲

Then, the '+' or X-basis states can be defined.

$$\begin{aligned} \triangleleft_+ &= \frac{1}{\sqrt{2}} (\triangleleft_0 + \triangleleft_1) \\ \triangleleft_- &= \frac{1}{\sqrt{2}} (\triangleleft_0 - \triangleleft_1) \end{aligned}$$

Along with the Z-basis states, these states form the building blocks of understanding quantum computing and the use of ZX-calculus for diagrammatic reasoning.

Intuitively, using the diagrammatic representation, we can consider combining diagrams in parallel. In the literature, this is the same as taking a *tensor product*. For example, we can have a 2-qubit state.

$$\triangleleft_{10} \text{---} = \triangleleft_{10} \text{---}$$

States can be transformed by quantum gates simply by *composing* them to obtain a new state. Given that composing the quantum gate U to a state $|\psi\rangle$ gives a state $|\psi'\rangle$, we can just connect the wires.

$$\triangleleft_{\psi'} \text{---} = \triangleleft_\psi \text{---} \square_U \text{---}$$

Intuitively, this means that the wire by itself should just be the identity gate, since composing it with any state should not change that state.

$$\begin{array}{c} \text{---} \boxed{I} \text{---} := \text{---} \\ \langle \psi | \text{---} \boxed{I} \text{---} = \langle \psi | \text{---} = \langle \psi | \end{array}$$

The dual of states are the effects. Here, we have the Z-basis effects.

$$\text{---} \triangleright_0 \quad \text{---} \triangleright_1$$

For a state $|\psi\rangle$ and an effect $\langle\phi|$, we can obtain a scalar inner product. This is essentially the *bra-ket* notation in diagrammatic form by just connecting the wires, which comes with a few other definitions.

Definition 1.2.3 (Inner product [2]). The *inner product* $\langle\phi|\psi\rangle$ of two states $|\psi\rangle$ and $|\phi\rangle$ is defined as

$$\langle \psi | \text{---} \triangleright_\phi$$

They are *orthogonal* if

$$\langle \psi | \text{---} \triangleright_\phi = 0$$

The *squared-norm* of a state $|\psi\rangle$ is

$$\langle \psi | \text{---} \triangleright_\psi$$

$|\psi\rangle$ is *normalised* if

$$\langle \psi | \text{---} \triangleright_\psi = \boxed{} = 1$$

where the empty diagram $\boxed{}$ represents the scalar 1. ▲

We can then define the inner product of the Z-basis states.

Definition 1.2.4. The inner product of the Z-basis states is

$$\langle 0 | \text{---} \triangleright_0 = \langle 1 | \text{---} \triangleright_1 = \boxed{}$$

$$\triangleleft 1 \longrightarrow 0 \triangleright = \triangleleft 0 \longrightarrow 1 \triangleright = 0$$

Succinctly, for $i, j \in \{0, 1\}$,

$$\triangleleft i \longrightarrow j \triangleright = \delta_{ij} = \begin{cases} \boxed{} & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

where δ_{ij} is the Kronecker delta. ▲

In fact, the succinct definition can be used to determine if a basis is an *orthonormal basis* (ONB). The X-basis is also an ONB.

We are almost at the stage where we can obtain probabilities from the diagrams. The scalar inner product is not really a probability, that is, a number between 0 and 1, but instead, a number $z \in \mathbb{C}$.

First, we note that putting multiple scalars in a single diagram is just the same as multiplying them together.

$$\begin{array}{c} \triangleleft \psi \longrightarrow \phi \triangleright \\ \triangleleft \psi' \longrightarrow \phi' \triangleright \end{array} = \triangleleft \psi \longrightarrow \phi \triangleright \cdot \triangleleft \psi' \longrightarrow \phi' \triangleright$$

We also know that for a complex number and its conjugate, $z, \bar{z} \in \mathbb{C}, |z|^2 = z\bar{z}$. Translating that to the diagrammatic representation, we can obtain the following.

$$\left| \triangleleft \psi \longrightarrow \phi \triangleright \right|^2 = \begin{array}{c} \triangleleft \psi \longrightarrow \phi \triangleright \\ \triangleleft \psi \longrightarrow \phi \triangleright \end{array}$$

Note that assymetry is introduced to denote the conjugation. Remember that a complex number $z = a + bi \in \mathbb{C}$ has a conjugate $\bar{z} = a - bi \in \mathbb{C}$, so this corresponds to a vertical reflection in the diagram above.

Using the *Born rule*, if we make sure that $|\psi\rangle$ and $|\phi\rangle$ are normalised, we can obtain the following.

$$0 \leq \begin{array}{c} \triangleleft \psi \longrightarrow \phi \triangleright \\ \triangleleft \psi \longrightarrow \phi \triangleright \end{array} \leq 1$$

This can be interpreted as our desired probability - the probability of measuring $|\psi\rangle$ in the state $|\phi\rangle$.

We now have all we need to introduce the ZX-calculus notation.

1.3 ZX-Calculus

So far, we have introduced the representation of quantum states, effects, and inner products in a diagrammatic form. However, there still isn't much we can do without a way to manipulate them. In the previous section, we mentioned that we can apply quantum gates to states to obtain new states. Here, we will introduce the ZX-Calculus and show the highly intuitive representation of some quantum gates such as the Hadamard and CNOT gates, based on the building blocks from the previous section. The ZX-Calculus is a graphical calculus for quantum computations, developed by Coecke and Duncan in 2008 [3]. All operations are expressed as *spiders*, connected by *edges* (or *wires*).

1.3.1 Spiders

Definition 1.3.1 (Spiders).

$$\begin{aligned}
 \text{Green Spider } (\alpha) &:= \text{Configuration of } n \text{ triangles with } (0,0) \text{ and } (0,0) \\
 &\quad + e^{i\alpha} \text{ Configuration of } n \text{ triangles with } (1,1) \text{ and } (1,1) \\
 \text{Red Spider } (\alpha) &:= \text{Configuration of } n \text{ triangles with } (0,0) \text{ and } (1,1) \\
 &\quad + e^{i\alpha} \text{ Configuration of } n \text{ triangles with } (1,1) \text{ and } (0,0)
 \end{aligned}$$

▲

Here, we have green and red spiders, also known as Z and X spiders, made of the Z and X basis states respectively. Wires coming in from the left are *inputs*, and wires going out to the right are *outputs*. The α inside a spider node is also called

the *phase* of the spider. For example, with $\alpha = 0$, $e^{i\alpha} = \boxed{}$. It is convenient to denote spiders with $\alpha = 0$ as having no angle written inside the spider.

Immediately, we can see that the Z and X basis states can be formed from the definition of the spiders, up to the multiplicative constant $\sqrt{2}$. Note that $e^{i\pi} = -1$.

$$\begin{aligned}
 \textcircled{0} &= \triangleleft 0 \text{---} + \triangleleft 1 \text{---} = \sqrt{2} \triangleleft 0 \text{---} \\
 \textcircled{\pi} &= \triangleleft 0 \text{---} - \triangleleft 1 \text{---} = \sqrt{2} \triangleleft 1 \text{---} \\
 \textcircled{0} &= \triangleleft 0 \text{---} + \triangleleft 1 \text{---} = \sqrt{2} \triangleleft 0 \text{---} \\
 \textcircled{\pi} &= \triangleleft 0 \text{---} - \triangleleft 1 \text{---} = \sqrt{2} \triangleleft 1 \text{---}
 \end{aligned} \tag{1.1}$$

It turns out that in reasoning with spiders, we can ignore the multiplicative constants, since much of the manipulation of the diagrams is based on the structure of the diagrams themselves. The scalars are easy to determine using the definitions we have seen so far. By using $\approx$ to denote equality up to a non-zero factor, we have

$$\begin{aligned}
 \textcircled{0} &\approx \triangleleft 0 \text{---} & \textcircled{\pi} &\approx \triangleleft 1 \text{---} \\
 \textcircled{0} &\approx \triangleleft 0 \text{---} & \textcircled{\pi} &\approx \triangleleft 1 \text{---}
 \end{aligned} \tag{1.2}$$

Scalars can also be represented using just spiders.

$$\textcircled{\alpha} = 1 + e^{i\alpha} \quad \textcircled{\alpha} \textcircled{a\pi} = \sqrt{2} e^{ia\alpha}$$

We can also see the single-qubit gates in the ZX-Calculus.

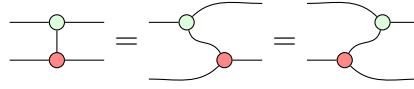
$$\begin{aligned}
 \text{---} \textcircled{\alpha} \text{---} &= \text{---} \triangleleft 0 \text{---} \triangleleft 0 \text{---} + e^{i\alpha} \text{---} \triangleleft 1 \text{---} \triangleleft 1 \text{---} = Z_\alpha \\
 \text{---} \textcircled{\alpha} \text{---} &= \text{---} \triangleleft 0 \text{---} \triangleleft 0 \text{---} + e^{i\alpha} \text{---} \triangleleft 1 \text{---} \triangleleft 1 \text{---} = X_\alpha
 \end{aligned}$$

We can then define many of the common quantum gates in this way.

$$\begin{aligned}
 I &= \text{---} \textcircled{0} \text{---} = \text{---} \textcircled{\pi} \text{---} = \text{---} \\
 X &= \text{---} \textcircled{\pi} \text{---} \\
 Z &= \text{---} \textcircled{\pi} \text{---} \\
 H &= e^{-i\frac{\pi}{4}} \cdot \text{---} \textcircled{\frac{\pi}{2}} \textcircled{\frac{\pi}{2}} \textcircled{\frac{\pi}{2}} \text{---} =: \text{---} \textcircled{\frac{\pi}{2}} \text{---} \equiv \text{---} \text{---} \text{---} \\
 S &= \text{---} \textcircled{\frac{\pi}{2}} \text{---} \\
 T &= \text{---} \textcircled{\frac{\pi}{4}} \text{---} \\
 CNOT &= \sqrt{2} \cdot \text{---} \textcircled{0} \text{---}
 \end{aligned} \tag{1.3}$$

There are 2 gates of special note here. The Hadamard gate H , denoted by the small yellow box, or as a blue dashed line, is a single-qubit gate that interchanges Z and X bases. We will see later in the ZX-rules that this is vital for the colour change (cc) rule, as well as why it is useful to denote it as the blue dashed line.

The CNOT gate is a two-qubit gate that acts on the target qubit if the control qubit is in the state $|1\rangle \approx \text{red circle with } \pi$. The vertical line is used simply because it does not matter which direction the line is going, where the following equality of the second and third forms can be verified.



Last but not least, we also introduce the SWAP gate, which as the name suggests, swaps two qubits.

$$SWAP = \text{crossing} := \sum_{i,j \in \{0,1\}} \begin{array}{cc} \text{triangle } j & \text{triangle } i \\ \text{triangle } i & \text{triangle } j \end{array}$$

We can now introduce the ZX-rules in the next section.

1.3.2 ZX-Rules

The ZX-Calculus is based on a set of rules that allow us to manipulate the diagrams.

For $\alpha, \beta \in [0, 2\pi)$ and $a \in \{0, 1\}$, the rules are as follows.

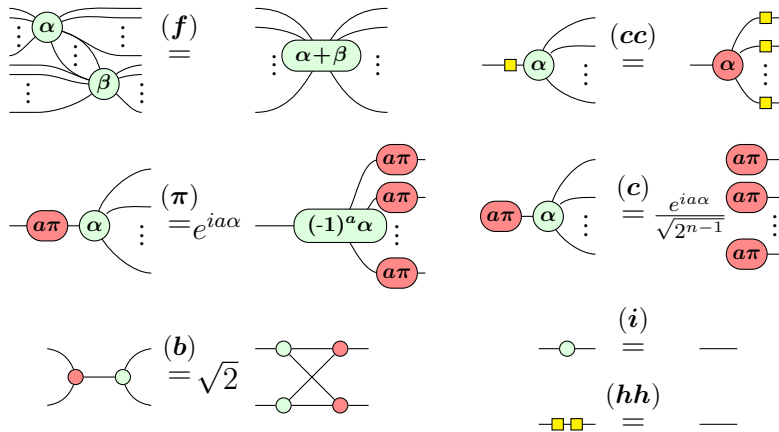


Figure 1.1: ZX-rules presented in [4]

The names of the rules are spider (**f**)usion, colour change (**cc**) rule, (**π**)-commutation, (**c**)opy rule, (**b**)ialgebra, (**i**)dentity rule, and (**hh**)adamard cancel rule. For (**c**), n refers to the number of outputs. By duality, the rules also hold if the colours are swapped.

The power of ZX-diagrams comes from the fact that two diagrams are equivalent as long as the connectivity of the diagram is the same - that is, as long as order of inputs and outputs of the whole diagram is preserved. This does not depend on the position of the spiders or the direction of wires between them. This neat feature of ZX-diagrams is called "Only Connectivity Matters" (OCM).

We can then obtain caps and cups using 2-legged spiders together with (**i**).

$$\bigcup \stackrel{(i)}{=} \bigcup \quad \bigcap \stackrel{(i)}{=} \bigcap$$

Caps and cups obey the *yanking equations*.

$$\text{---} = \text{---} \quad \text{---} = \text{---}$$

Before we move on to reasoning about the diagrams in a graph-theoretic manner, we first introduce the concept of a *Clifford+T diagram*.

Definition 1.3.2 (Clifford diagram). A *Clifford diagram* is a ZX-diagram where the phases are restricted to integer multiples of $\frac{\pi}{2}$. ▲

Definition 1.3.3 (Clifford+T diagram). A *Clifford+T diagram* is a ZX-diagram where the phases are restricted to integer multiples of $\frac{\pi}{4}$. ▲

It is clear from the definitions that a Clifford diagram is a subset of a Clifford+T diagram. The reason we introduce these concepts is that in trying to obtain our main goal of the thesis, it is sufficient to reason about Clifford+T diagrams, as a result of the following theorem.

Theorem 1.3.4 (Approximate Universality [5]). Clifford+T diagrams are *approximately universal* for quantum computing - they can approximate any quantum circuit.

The proof of the above theorem involves the *Solovay-Kitaev theorem*, combining with the fact that any single-qubit phase gate can be approximated by a sequence of Hadamard and T gates, with details found in [5].

1.4 Graph Reduction

Because of OCM, ZX-diagrams up until now have very similar structure to graphs, where the spiders are the vertices and the wires are the edges. The only difference may be that the spiders are coloured, and that the Hadamard boxes are included. In this section, we solidify the connection between ZX-diagrams and graphs, and introduce an algorithm that will simplify ZX-diagrams, defining what it means for a graph to be simplified. Ultimately, simplifying ZX-diagrams should allow us to reduce the number of spiders in the diagram, which directly translates to a reduction in the number of gates in the quantum circuit. In turn, as we will see in the next chapter, this will lead to a speedup in the simulation of quantum circuits.

First, we define what it means for ZX-diagram to be *graph-like*.

Definition 1.4.1 (Graph-like [6]). A ZX-diagram is *graph-like* when

- Every spider is a Z-spider.
- Spiders are only connected via Hadamard edges.
- There is at most only one edge between each pair of spiders.
- There are no self-loops.
- Every input and output wire is connected to a Z-spider.

▲

The following proposition can then be proven, using rewrite rules introduced previously in ZX-Rules.

Proposition 1.4.2 ([6]). Every ZX-diagram can be converted to a graph-like ZX-diagram in polynomial time.

The proof [6] makes use of the fact that Hadamard edges in parallel cancel, and that we can always remove a Hadamard self-loop.

$$\begin{array}{c} \cdots \text{---} \alpha \text{---} \beta \text{---} \cdots \approx \cdots \text{---} \alpha \text{---} \beta \text{---} \cdots \\ \text{---} \alpha \text{---} \approx \text{---} \alpha + \pi \text{---} \end{array}$$

We can simplify the graph-like ZX-diagram further. Now, the following two rewrite rules, derivable from the original ZX-rules, are introduced.

$$\begin{array}{c} \begin{array}{ccc} \alpha_1 & & \alpha_n \\ & \searrow & \nearrow \\ & \pm \frac{\pi}{2} & \\ & \nearrow & \searrow \\ \alpha_2 & & \alpha_3 \end{array} & \stackrel{(LC)}{=} e^{\pm i \frac{\pi}{4}} \sqrt{2}^{\frac{(n-1)(n-2)}{2}} \begin{array}{ccc} \alpha_1 \mp \frac{\pi}{2} & & \alpha_n \mp \frac{\pi}{2} \\ & \searrow & \nearrow \\ & & \vdots \\ & \nearrow & \searrow \\ \alpha_2 \mp \frac{\pi}{2} & & \alpha_3 \mp \frac{\pi}{2} \end{array} \\ \\ \begin{array}{ccc} \alpha_1 & & \gamma_1 \\ & \searrow & \nearrow \\ & j\pi & k\pi \\ & \nearrow & \searrow \\ \alpha_n & & \gamma_l \\ & \searrow & \nearrow \\ & \beta_1 & \vdots \\ & \vdots & \vdots \\ & \beta_m & \end{array} & \stackrel{(P)}{=} (-1)^{jk} \sqrt{2}^E \begin{array}{ccc} \alpha_1 + k\pi & & \gamma_1 + j\pi \\ & \searrow & \nearrow \\ & & \vdots \\ & \nearrow & \searrow \\ \alpha_n + k\pi & & \gamma_l + j\pi \\ & \searrow & \nearrow \\ & \beta_1 + (j+k+1)\pi & \\ & \vdots & \\ & \beta_m + (j+k+1)\pi & \end{array} \end{array}$$

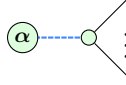
Figure 1.2: Derived ZX-rules [7], where $\alpha, \beta, \gamma \in [0, 2\pi)$ and $j, k \in \{0, 1\}$.

The rules are *local complementation* (**LC**) and *pivoting* (**P**) respectively. Here, $E = (n-1)m + (l-1)m + (n-1)(l-1)$.

Repeatedly simplifying and reducing diagrams in this manner will end up giving us patterns in the diagrams. The following definitions help us identify these patterns. First, we define interior and boundary spiders, and then phase gadgets.

Definition 1.4.3 ([8]). A *boundary spider* is a spider connected to an input or output. All other spiders are *internal spiders*. ▲

Definition 1.4.4 (Phase gadget [8]). A *phase gadget* is an arity-1 spider with angle α , connected via a Hadamard edge to a spider with no angle.



▲

These help us define the reduced gadget form.

Definition 1.4.5 (Reduced gadget form [8]). A graph-like ZX-diagram is in *reduced gadget form* when

- Every internal spider is a non-Clifford spider or part of a non-Clifford phase-gadget.
- Every phase-gadget has more than one target.
- No two phase-gadgets have the same set of targets.

▲

Example 1.4.6. The following diagram from [9] is in reduced gadget form. ▲

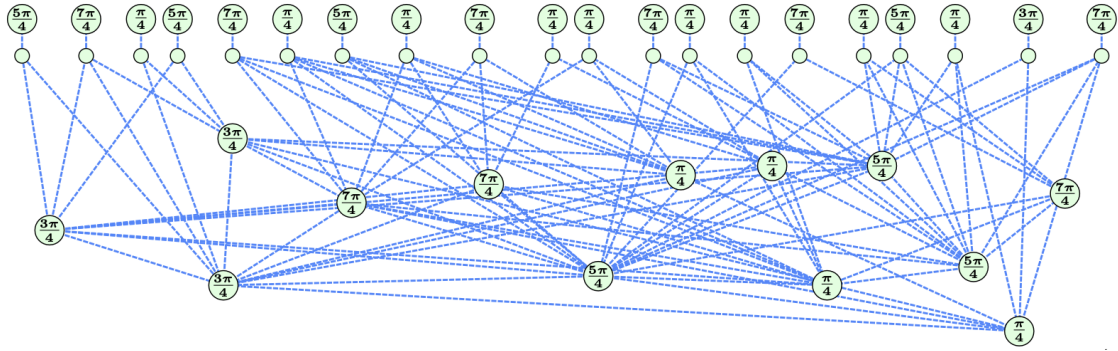


Figure 1.3: A ZX-diagram in reduced gadget form [9]

It turns out that there is an algorithm that is bounded in $O(n^3)$ where n is the number of spiders, to convert any graph-like ZX-diagram to reduced gadget form [9]. Thus, much of the work in this thesis will focus on the reduced gadget form, as it tends to bring us closer to the goal of reducing overall T -count. However, there are exceptions to this, as we will see in the next chapter.

2

Classical Simulation

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2.1 Introduction

As mentioned in the previous chapter, classical simulation of quantum circuits provides an essential tool for verification and validation, despite its inherent limitations. This chapter delves into the methods and advancements in classical simulation, particularly focusing on stabilizer decomposition and the role of magic states in extending the capabilities of these simulations. Classical simulation methods that store a complete description of an n -qubit quantum state as a complex vector of size 2^n are effective only for a small number of qubits, typically around 30. For example, Shor's factoring algorithm has been simulated with 31 qubits and approximately half a million gates (Paragraph 1). However, for certain classes of quantum circuits, such as those composed of gates from the Clifford group, classical simulation can be significantly more efficient. This is known as the *Gottesman-Knill theorem*.

Theorem 2.1.1 (Gottesman-Knill Theorem). A Clifford computation can be efficiently classically simulated.

Building on the foundational concepts discussed in the previous chapter, where we explored the graphical representation of quantum circuits using ZX-Calculus, we now turn to the specific techniques for simulating these circuits classically. ZX-Calculus not only aids in visualizing quantum operations but also simplifies the process of identifying and manipulating stabilizer states, which are central to many classical simulation methods (Paragraph 4).

Stabilizer circuits are particularly notable for their applications in quantum error correction. These circuits are initialized in stabilizer states, evolve under stabilizer operations, and are measured in the Pauli basis. Classical simulation of these circuits is efficient due to their restriction to the Clifford group operations (Paragraph 3). However, to achieve universal quantum computation, we must incorporate non-Clifford elements, such as magic states. The introduction of magic states, which extend the computational capabilities of stabilizer circuits, brings an exponential cost to the simulation. Nonetheless, when circuits are predominantly composed of Clifford gates, classical simulation remains feasible (Paragraph 3).

Appendices

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